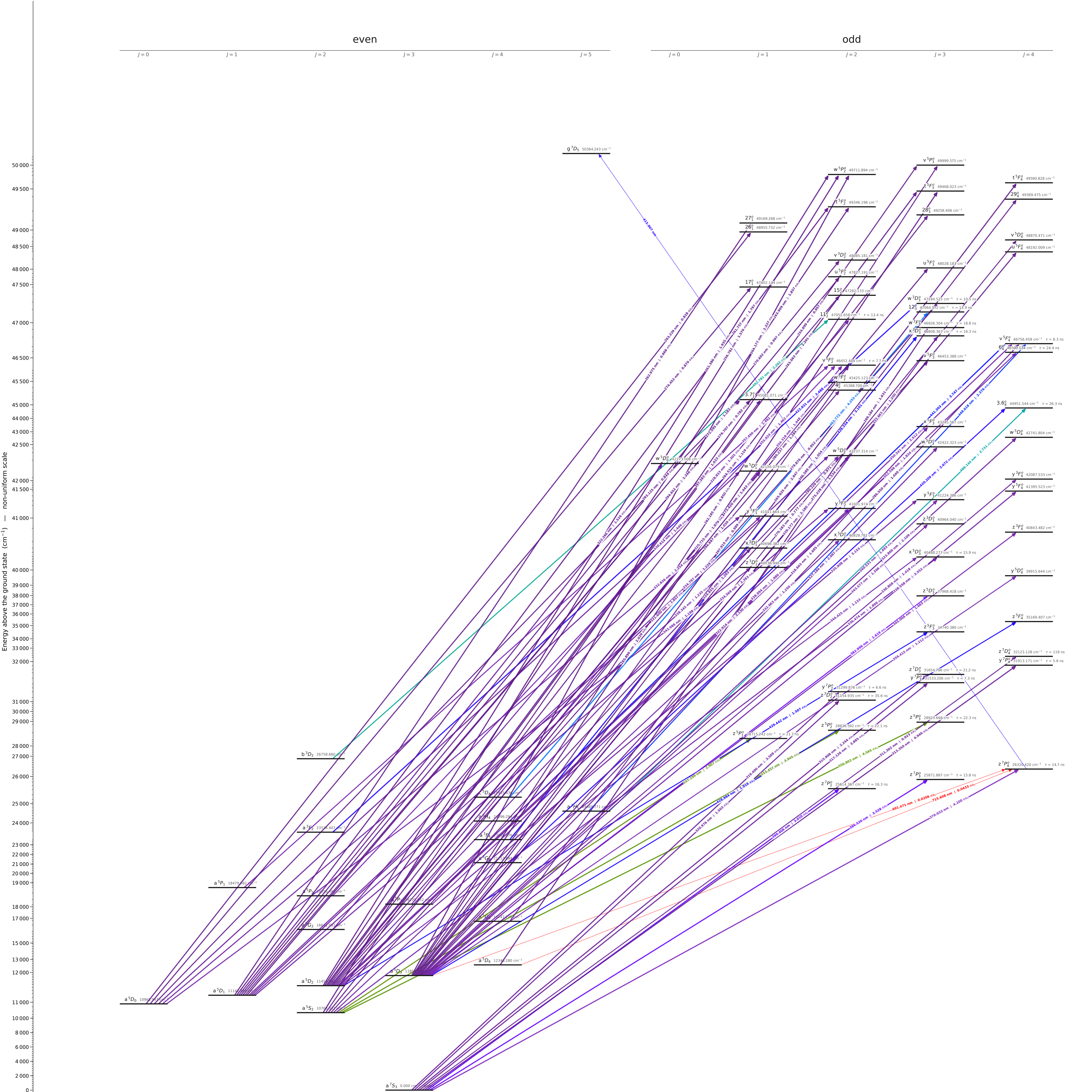
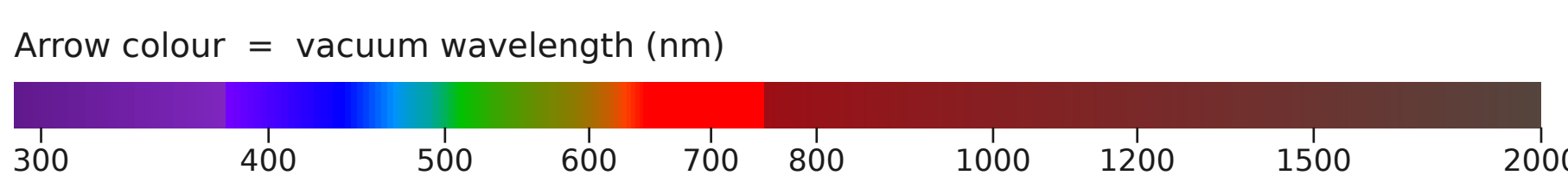


Molybdenum

Mo (Mo I). Strongest classified transitions below 2 μm (NIST): 82 levels, 90 lines.

Mo ⁺ ionisation limit 57204.30 cm⁻¹ = 7.09243 eV (174.812 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $[U|er|U']$ in ea.
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	[U er U'] (ea)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a ⁷ S ₃ – z ⁷ P ₁ ^o	379.9333	379.8255	789.0659(6)	4.100	6.900e+07	10.83	100	laser-cooling / MOT cycling transition
a ⁷ S ₃ – z ⁷ P ₂ ^o	386.5199	386.4104	775.6197	3.528	6.240e+07	10.07	98.6	member of the a ⁷ S ₃ - z ⁷ P resonance triplet
a ⁷ S ₃ – z ⁷ P ₂ ^o	390.4059	390.2953	767.8994	3.010	6.170e+07	9.764	100	member of the a ⁷ S ₃ - z ⁷ P resonance triplet
z ⁷ P ₁ ^o – g ⁷ D ₅	415.9072	415.7900	720.8157	–	–	–	–	second step of a two-colour, three-photon resonance-ioniz...
a ⁵ D ₃ – z ⁷ P ₂ ^o	691.4710	691.2803	433.5575	0.021	2.900e+02	10.83	0	second leak channel of the 380 nm cooling transition
a ⁵ D ₄ – z ⁷ P ₂ ^o	715.6084	715.4112	418.9367(28)	0.043	1.150e+03	10.83	0	MOT repump line

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).

⁹⁵Mo hyperfine structure (I = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
a ⁵ S ₃	-208.582060(10)	0.03705(10)	F=1/2: +2085.84 F=3/2: +1772.96 F=5/2: +1251.49 F=7/2: +521.44 F=9/2: -417.18 F=11/2: -1564.36

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
a ⁷ S ₃ – z ⁷ P ₁ ^o	379.933	92-96	920.4(19)
a ⁷ S ₃ – z ⁷ P ₂ ^o	379.933	94-96	418.4(16)
a ⁷ S ₃ – z ⁷ P ₂ ^o	379.933	98-96	-289.4(16)
a ⁷ S ₃ – z ⁷ P ₂ ^o	379.933	100-96	-812.0(19)

Shift = ν(first isotope) – ν(second isotope).